Computer Vision and Multiview Geometry DT084A Lab - 6 Report

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Task: Pose Estimation Algorithms

Pose estimation is the problem of estimating the position and orientation of a camera with respect to a set of known 3D points. In classical computer vision, this problem is commonly formulated as the Perspective-n-Point (PnP) problem.

Given a set of n 3D points in the world coordinate system and their corresponding 2D projections in the image plane, the goal is to estimate the camera pose consisting of a rotation matrix R and a translation vector t.

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \, [R \mid t] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

where K is the camera intrinsic matrix, (X, Y, Z) are the 3D coordinates, and (u, v) are the image coordinates.

Algorithm	Advantages	Disadvantages
P3P (Perspective-3-Point)	Very fast; minimal solution using only 3 points	Multiple possible solutions; sensitive to noise
EPnP (Efficient PnP)	Computationally efficient; works with many points; widely used in real-time systems	Less robust when points are noisy or poorly distributed
DLS (Direct Least Squares)	Provides accurate pose estimation; handles larger datasets well	Computationally more expensive
UPnP (Universal PnP)	Can estimate pose even with unknown focal length	Less stable when image noise is present
RPnP (Robust PnP)	More robust against noise; high accuracy	Slightly higher computational cost
Iterative PnP (Levenberg–Marquardt refinement)	Very accurate after initialization	Requires good initial pose estimation

Table 1: Comparison of classical PnP pose estimation algorithms

Other Pose Estimation Algorithms

Besides PnP-based methods, several other approaches can be used for camera pose estimation.

- **Direct Linear Transformation (DLT)** A classical method used to estimate camera pose from point correspondences using linear algebra. It is simple but sensitive to noise.
- **Iterative Closest Point (ICP)** Commonly used when aligning two 3D point clouds. ICP estimates the rigid transformation between them and can therefore recover pose.
- **Essential Matrix Decomposition** Used in stereo vision or multi-view geometry. The relative pose between two cameras can be recovered by decomposing the essential matrix obtained from feature correspondences.
- **Homography-based Pose Estimation** When the observed scene is planar, camera pose can be estimated from a homography transformation between the image plane and the world plane.

These methods are commonly used in applications such as visual SLAM, augmented reality, robotics, and 3D reconstruction.

References

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Task 2: 2D Transformations

A rectangle is defined by the corners

$$(0,0),\ (0,3),\ (5,3),\ (5,0).$$

1. Similarity Transform Matrix

The rectangle is translated 3 units horizontally, 2 units vertically, and rotated by 15° clockwise.

In homogeneous coordinates, a 2D similarity transform is written as

$$T = \begin{bmatrix} s\cos\theta & -s\sin\theta & t_x \\ s\sin\theta & s\cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix}.$$

Since no scaling is specified, we take $s = 1$. A clockwise rotation of 15° corresponds to $\theta = -15°$. Therefore,

$$T = \begin{bmatrix} \cos(-15°) & -\sin(-15°) & 3 \\ \sin(-15°) & \cos(-15°) & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Using

$$\cos(-15°) = \cos(15°), \qquad \sin(-15°) = -\sin(15°),$$

the matrix becomes

$$T = \begin{bmatrix} \cos 15° & \sin 15° & 3 \\ -\sin 15° & \cos 15° & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Numerically,

$$T \approx \begin{bmatrix} 0.9659 & 0.2588 & 3 \\ -0.2588 & 0.9659 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

2. Transformation Matrix for the New Corner Positions

The rectangle is transformed so that its corners become

$$(1, 1),\ (3, 3),\ (6, 3),\ (5, 2).$$

Thus, the point correspondences are

$$(0, 0) \to (1, 1),$$
$$(0, 3) \to (3, 3),$$
$$(5, 3) \to (6, 3),$$
$$(5, 0) \to (5, 2).$$

The homography matrix is written as

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix},$$

where the transformed point is obtained by

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \qquad \left(\frac{x'}{w'}, \frac{y'}{w'}\right).$$

Derivation of the Transformation Matrix

A projective transformation (homography) between two planes is written as

$$H = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & 1 \end{bmatrix}$$

where the transformed coordinates are obtained as

$$x' = \frac{ax + by + c}{gx + hy + 1}, \qquad y' = \frac{dx + ey + f}{gx + hy + 1}.$$

Using the first correspondence

$$(0, 0) \to (1, 1)$$

we obtain

$$1 = \frac{c}{1}, \qquad 1 = \frac{f}{1}$$

which gives

$$c = 1, \qquad f = 1.$$

Using the second correspondence

$$(0, 3) \to (3, 3)$$

we obtain

$$3 = \frac{3b + 1}{3h + 1}, \qquad 3 = \frac{3e + 1}{3h + 1}.$$

Similarly, the remaining correspondences

$$(5, 3) \to (6, 3), \qquad (5, 0) \to (5, 2)$$

produce additional equations for the unknown parameters. Solving the resulting linear system yields

$$g = \frac{1}{5}, \qquad h = 0.$$

Substituting these values back into the previous equations gives

$$a = \frac{9}{5}, \quad b = \frac{2}{3}, \quad d = \frac{3}{5}, \quad e = \frac{2}{3}.$$

Thus the homography matrix becomes

$$H = \begin{bmatrix} \frac{9}{5} & \frac{2}{3} & 1 \\ \frac{3}{5} & \frac{2}{3} & 1 \\ \frac{1}{5} & 0 & 1 \end{bmatrix}.$$

This matrix maps the original rectangle corners exactly to the required target coordinates. Using the four point correspondences, the resulting homography is

$$H = \begin{bmatrix} \frac{9}{5} & \frac{2}{3} & 1 \\ \frac{3}{5} & \frac{2}{3} & 1 \\ \frac{1}{5} & 0 & 1 \end{bmatrix}.$$

Its numerical form is

$$H \approx \begin{bmatrix} 1.8 & 0.6667 & 1 \\ 0.6 & 0.6667 & 1 \\ 0.2 & 0 & 1 \end{bmatrix}.$$

This matrix maps the original rectangle corners exactly to the required new positions:

$$(0,0) \mapsto (1,1), \quad (0,3) \mapsto (3,3), \quad (5,3) \mapsto (6,3), \quad (5,0) \mapsto (5,2).$$

Task 3: Image Stitching

The stitching pipeline consisted of the following steps:

1. Detect feature points and descriptors using SIFT.
2. Match the descriptors between the two images using brute-force matching and Lowe's ratio test.
3. Remove incorrect correspondences using RANSAC.
4. Estimate the homography from the remaining inlier correspondences using the DLT formulation solved by SVD.
5. Warp the first image into the plane of the second image using perspective warping.
6. Combine the warped image and the second image into a panorama.

The homography was estimated in homogeneous coordinates. For corresponding points

$$\mathbf{x} = [x, y, 1]^T \quad \text{and} \quad \mathbf{x}' = [x', y', 1]^T,$$

the mapping is written as

$$\mathbf{x}' \sim H\mathbf{x},$$

where H is the 3×3 homography matrix. The matrix was computed using the Direct Linear Transformation (DLT) method and solved with Singular Value Decomposition (SVD). To improve robustness against mismatched correspondences, RANSAC was applied before the final homography estimation.

For warping, perspective warping was used. In practice, inverse warping is preferred because it maps destination pixels back to the source image and reduces the holes that typically occur in forward warping.

Feature Matching Results

Fig 1 shows the initial feature matches obtained after SIFT detection and descriptor matching.

Figure 1: Initial feature matches obtained from SIFT descriptors.

The raw matches indicate that many corresponding points were successfully detected on repeated structures such as the window frames, building windows, roof edges, and decorative lights. Most of the match lines are approximately horizontal, which is consistent with the fact that the second image was captured by shifting the viewing direction mainly along the horizontal axis. However, not all of these matches are equally reliable, and some of them may still contain outliers. Fig 2 shows the inlier correspondences selected by RANSAC.

Figure 2: Inlier feature matches after RANSAC filtering.

After RANSAC, only the matches that are consistent with a single homography remain. Compared to the raw matches, the remaining correspondences are more geometrically consistent. This step is essential because homography estimation is highly sensitive to incorrect matches, and even a small number of outliers can significantly degrade the final stitching result.

Using the inlier correspondences, a homography was estimated and the first image was warped into the coordinate system of the second image. The result is shown in Figure 3.

The warped image appears inside a larger black canvas because the transformed image no longer occupies a rectangular region after the perspective transformation. The black

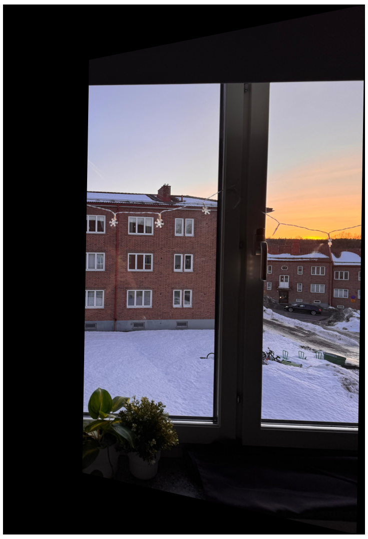

Figure 3: First image warped into the coordinate system of the second image.

regions correspond to empty pixels where no source values are available. The final stitched panorama is shown in Fig 4.

Figure 4: Final stitched panorama.

However, the result is not perfectly seamless. This is likely caused by several factors:

- the camera center was not perfectly fixed during capture,
- the scene is not fully planar,
- the window frame is much closer to the camera than the building outside,
- the two images have noticeable brightness and exposure differences.

These effects are typical in homography-based stitching. A single homography can align one dominant plane or approximate a pure camera rotation, but it cannot perfectly align all scene points when strong depth differences are present.

The final panorama is visually reasonable and demonstrates that the stitching pipeline works correctly.